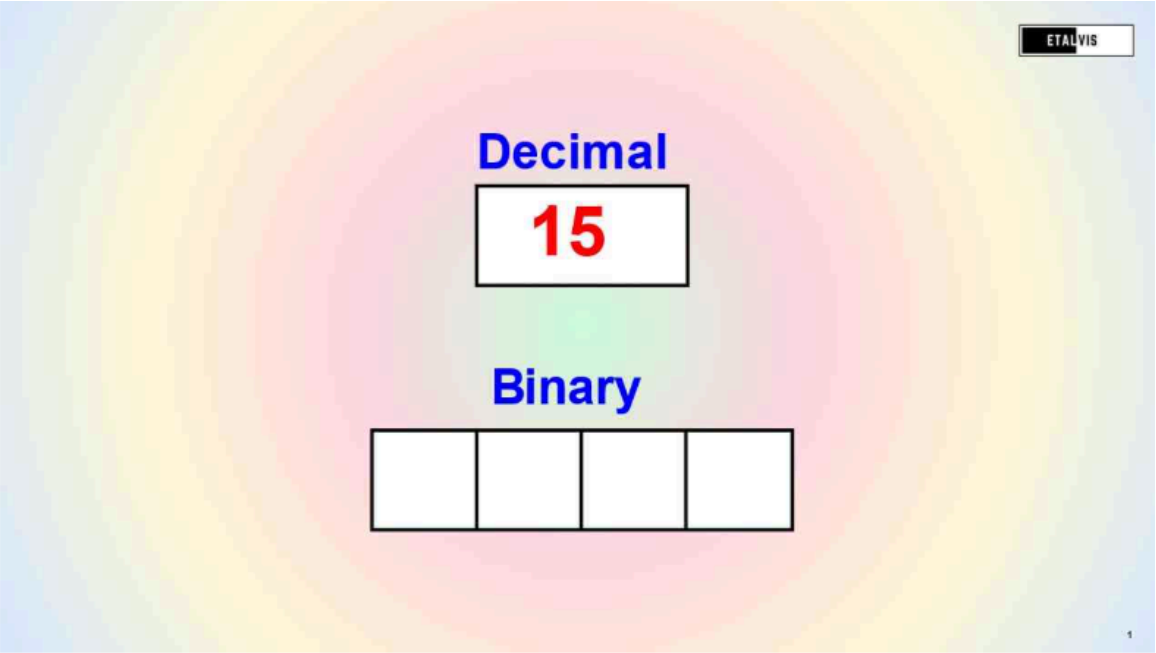
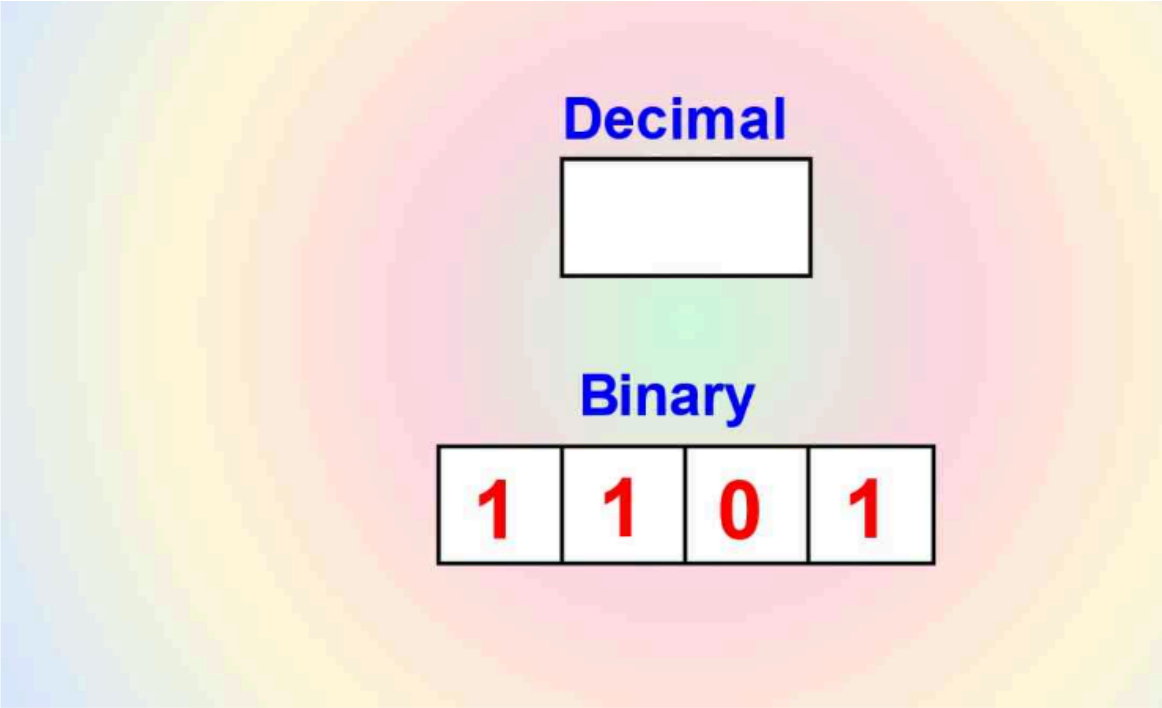
Number Systems

Assessment:1



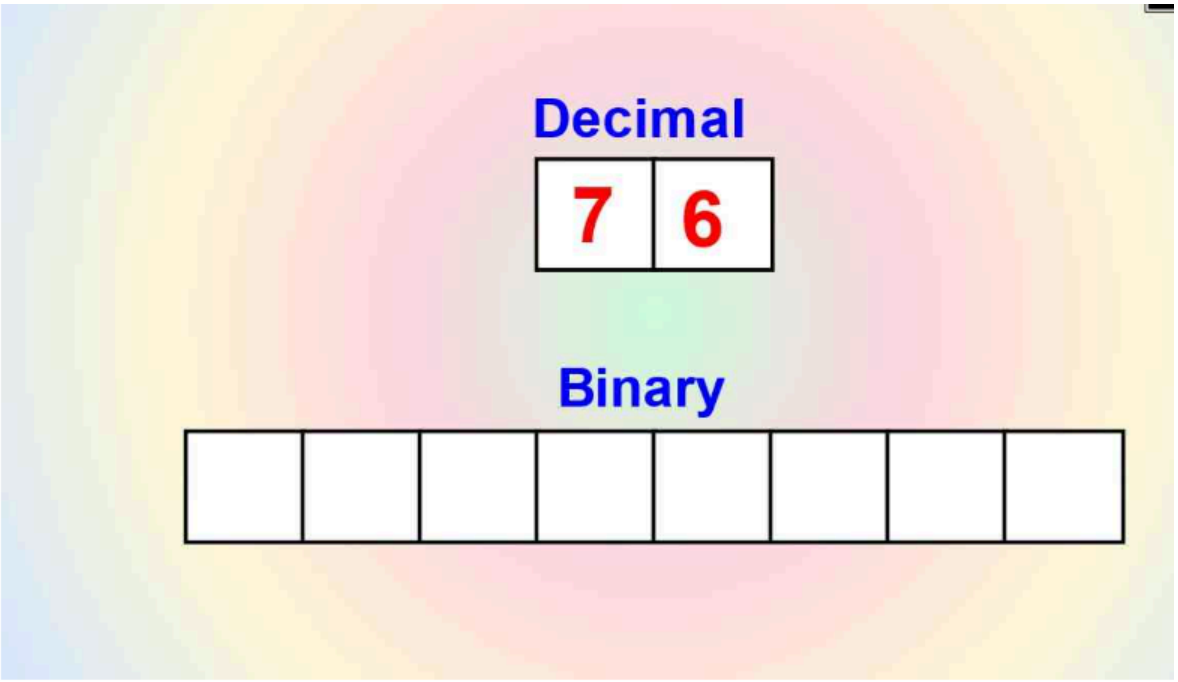
What is the Binary value?

1001	1111
1100	0011



What is the decimal value?

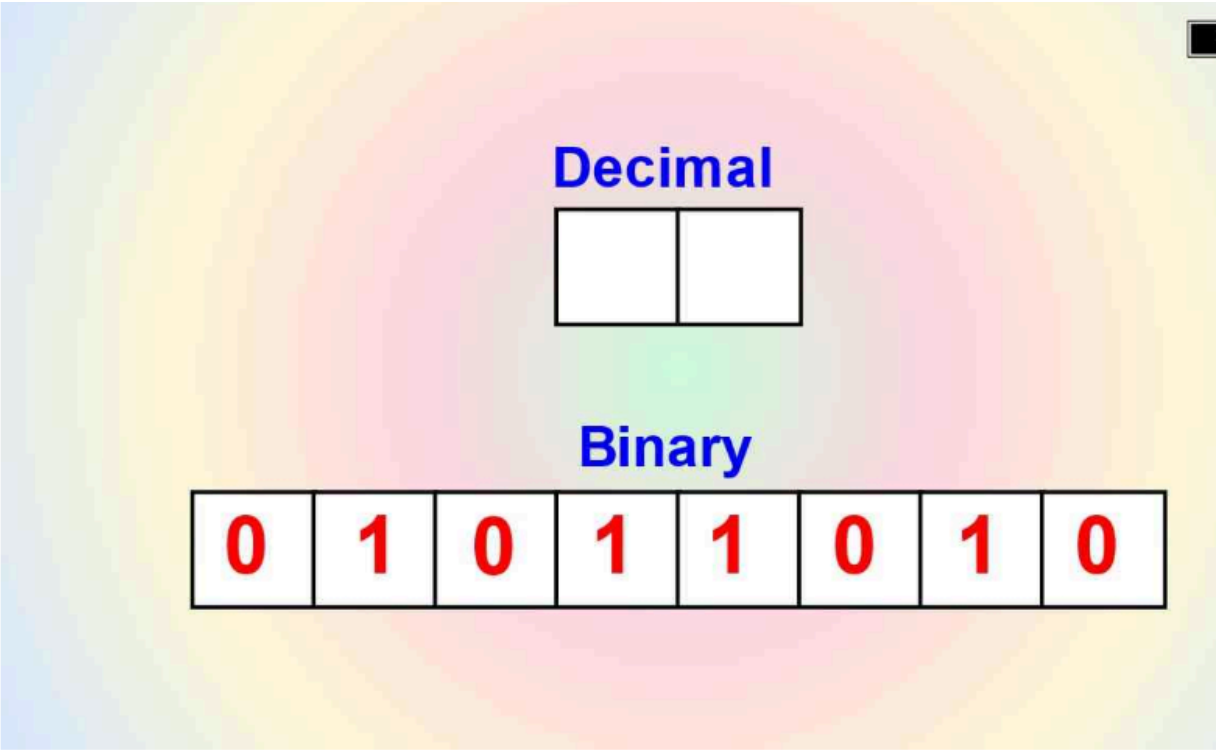
- 11
- 15
- 12
- 13



What is the binary value?

- 01001110 ✕
- 01001100
- 10001100
- 00101100

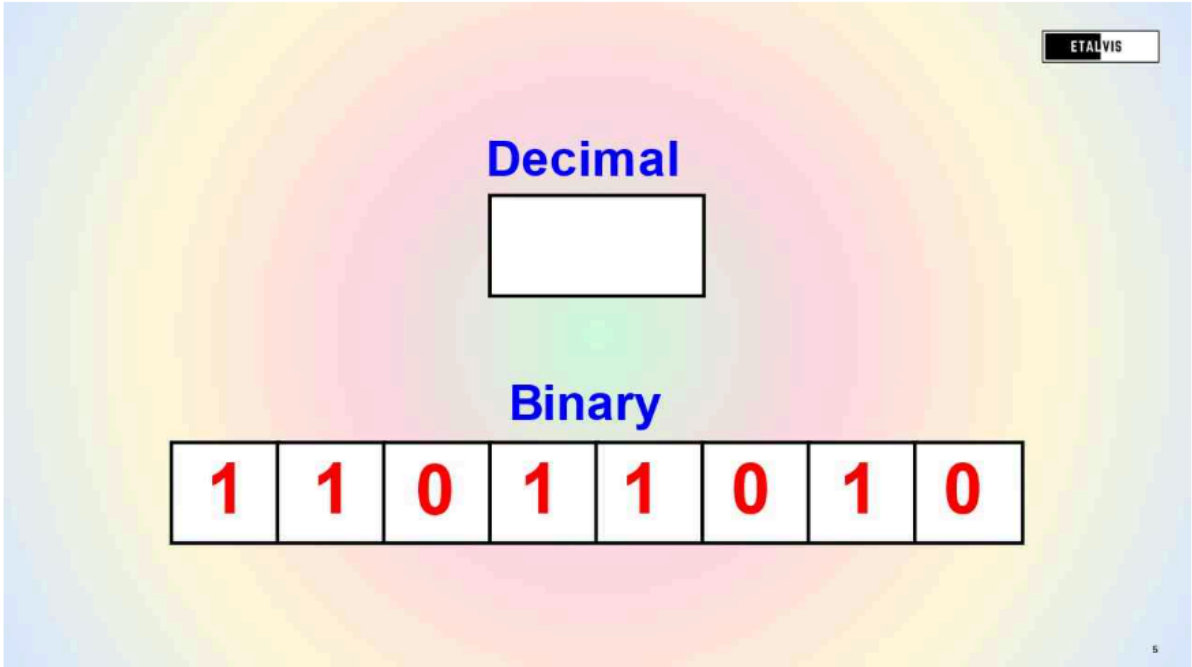
4.



What is the decimal value?

92	88
90 ✓	160

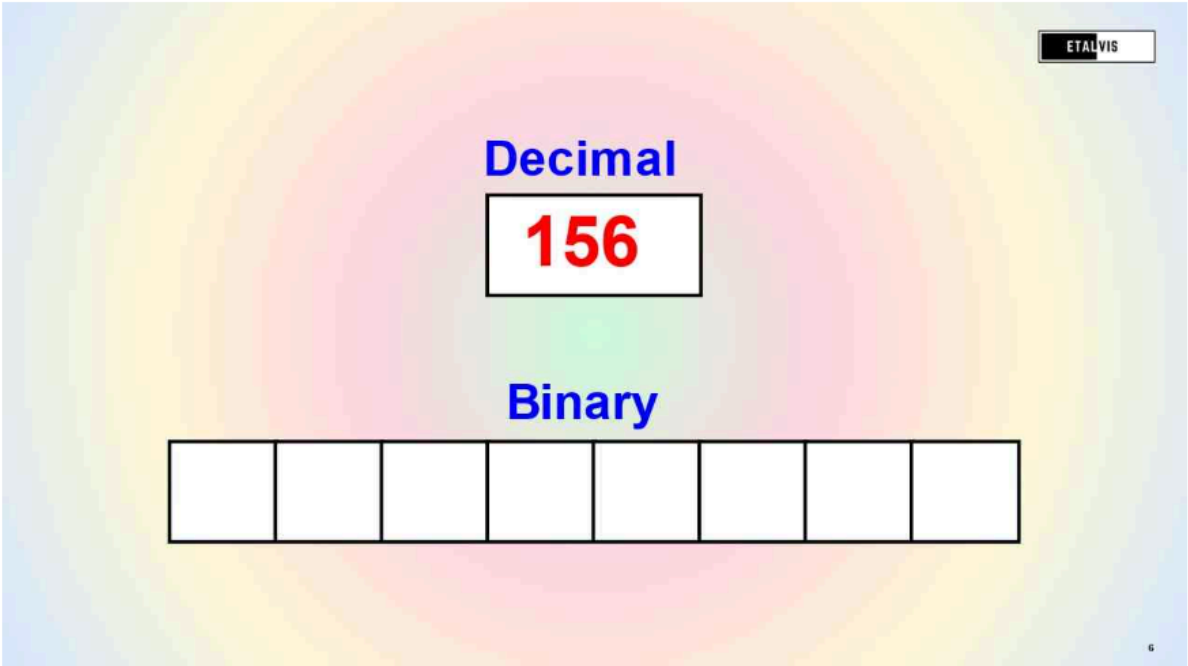
5.



What is the Decimal Value?

- 218 ✓
- 136
- 98
- 216

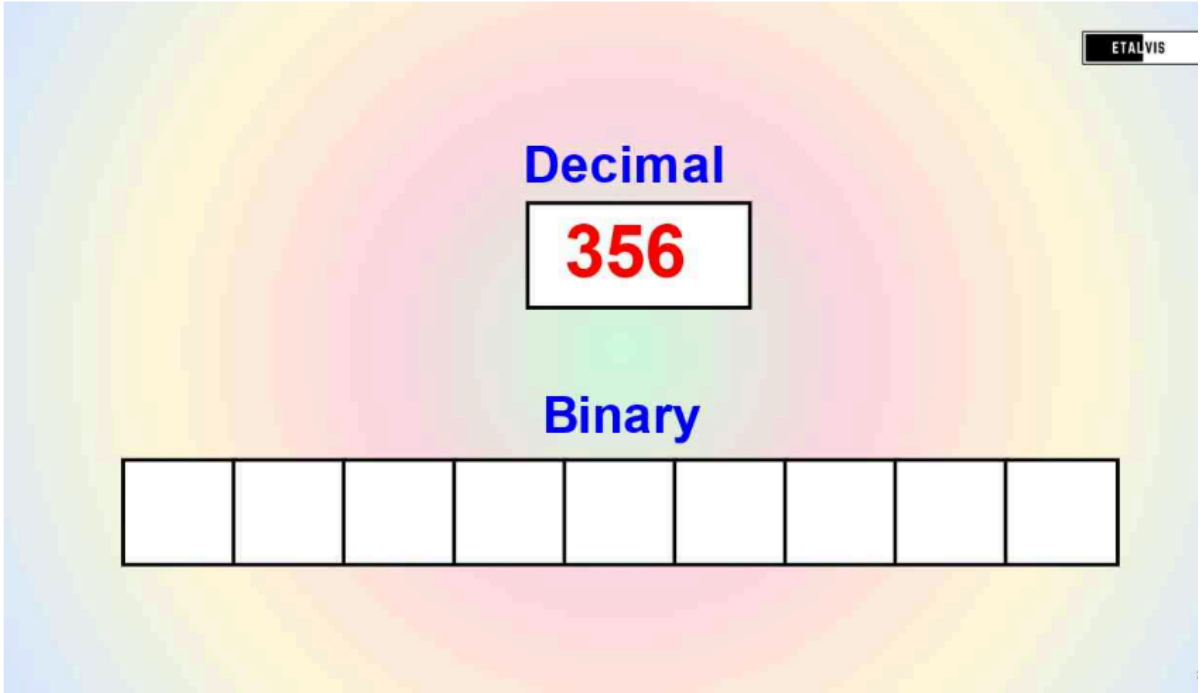
6.



What is the binary value?

- 01001110
- 10011100
- 10001110
- 00111110

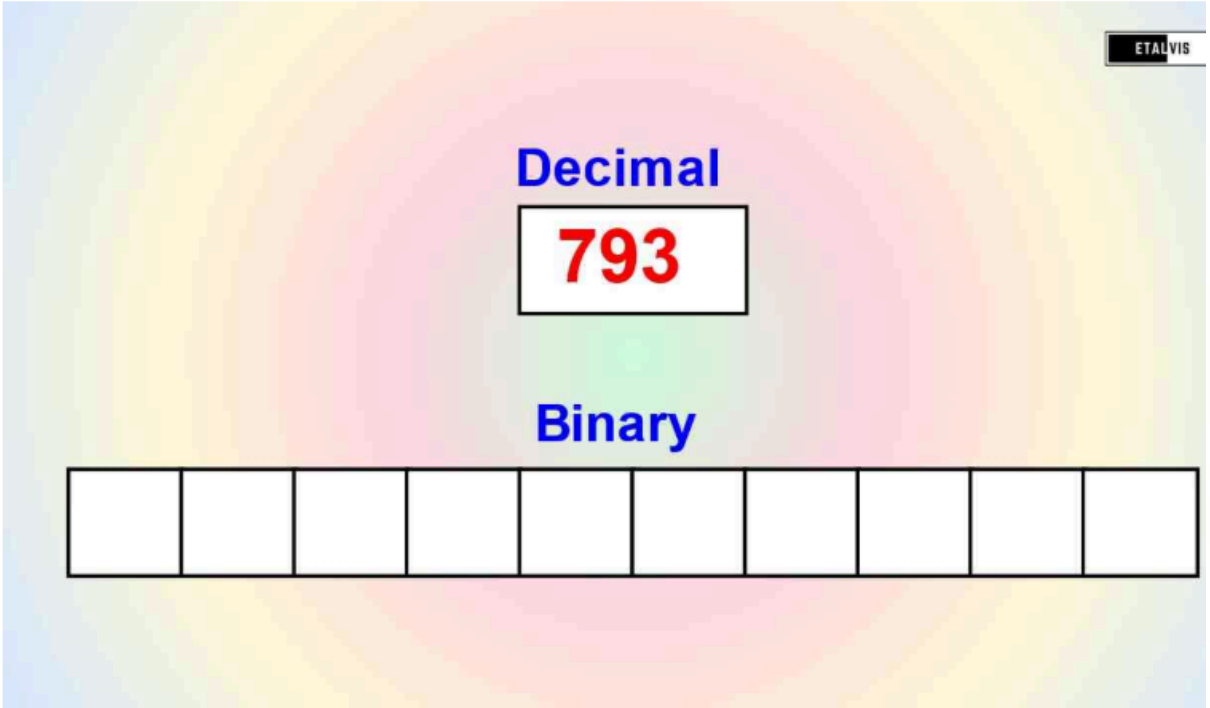
7.



What is the Binary Value?

- 11001100
- 111001100
- 101100100 ✓
- 01100110

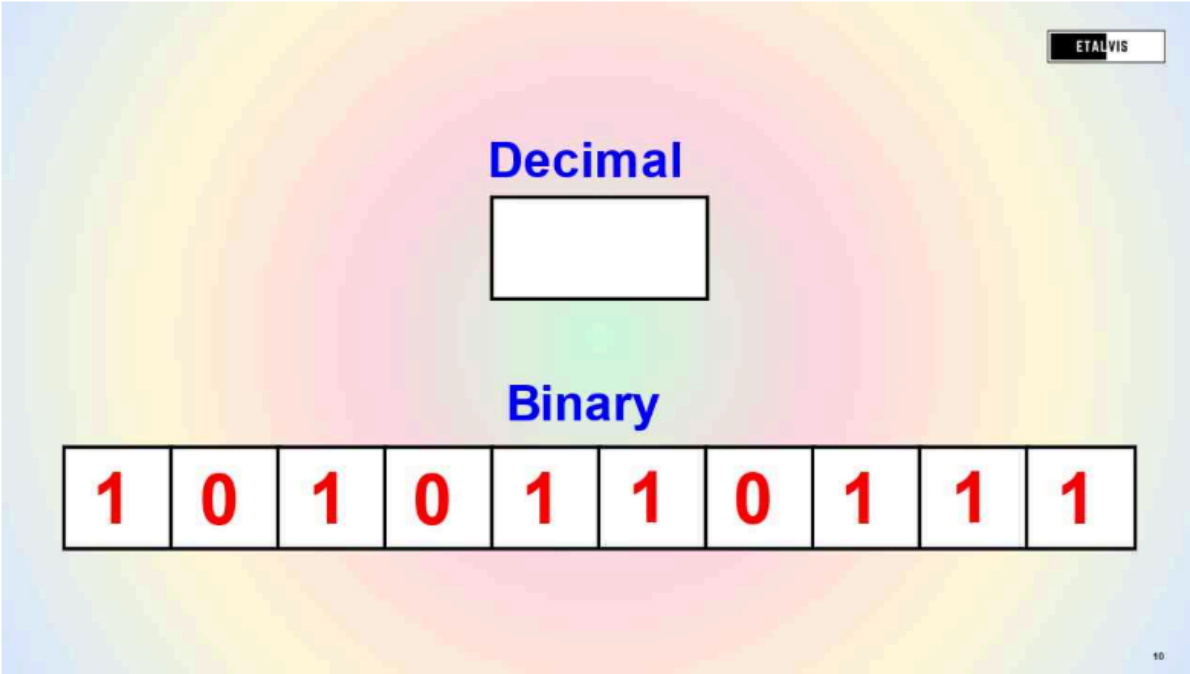
9.



What is the BInary Value?

- 11101110
- 11001110
- 111001110
- 1100011001

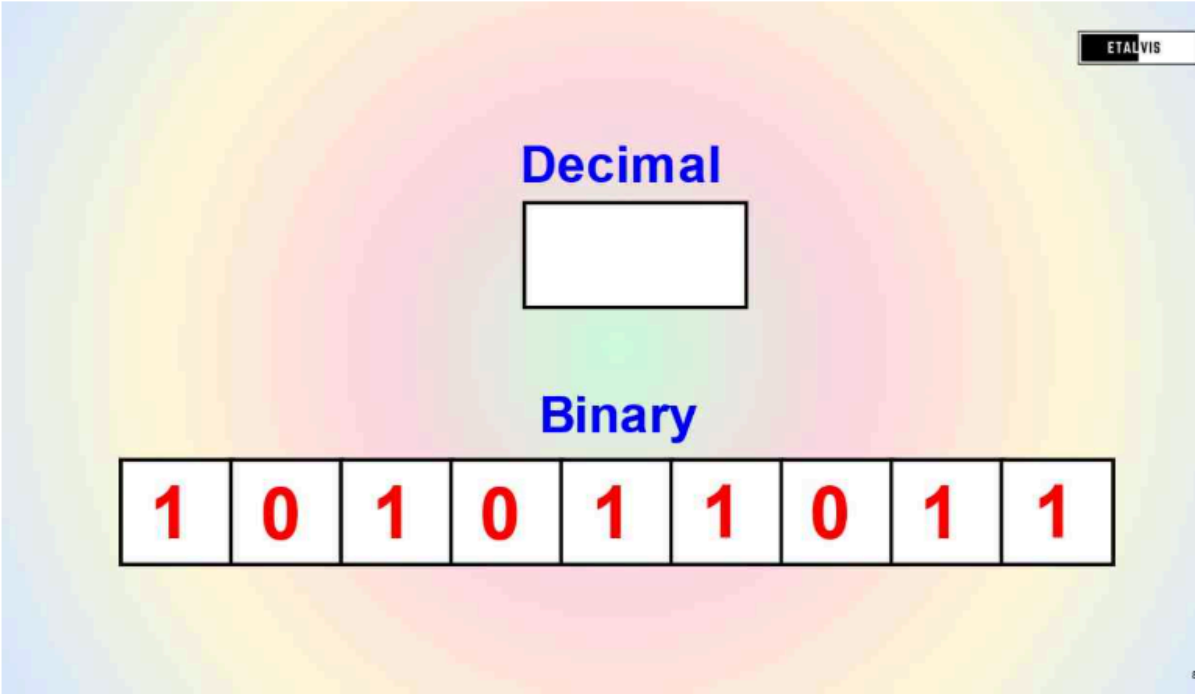
10.



What is the Decimal Value

- 595
- 695
- 295
- 395

8.

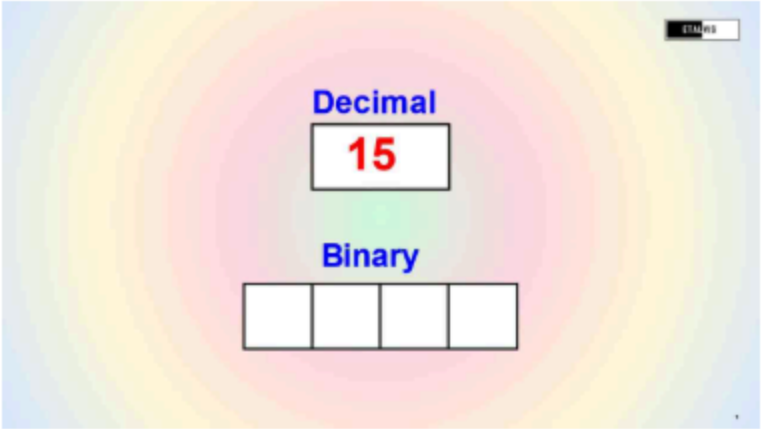


What is the Decimal Value?

- 347 ✓
- 357
- 256
- 197

Assessment:2

1.

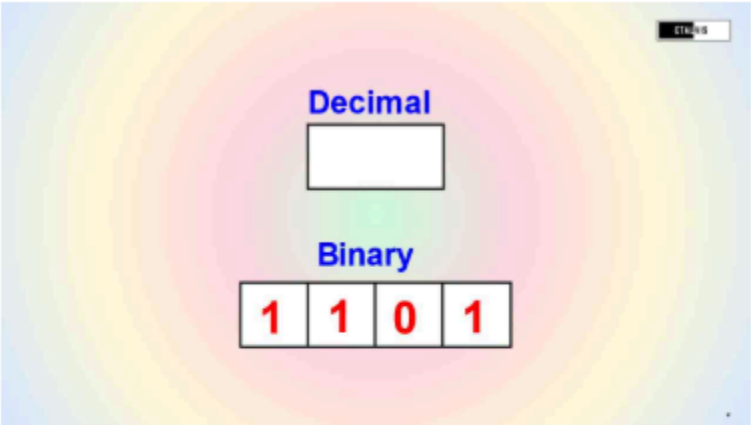


Marks: 1

What is the Binary value?

- 1001
- 1111 ✓
- 1100
- 0011

2.

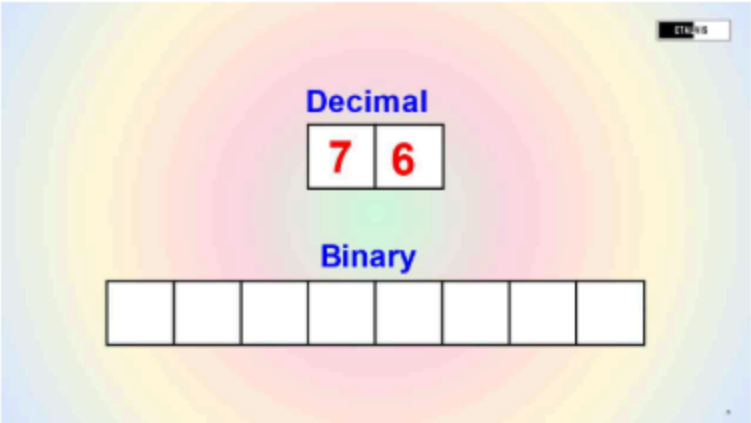


Marks: 1

What is the decimal value?

- 11
- 15
- 12
- 13 ✓

3.

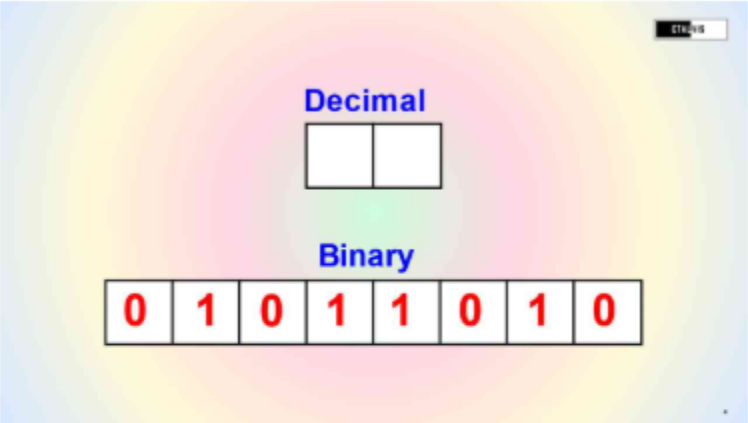


Marks: 1

What is the binary value?

- 01001110 ✗
- 01001100 ✓
- 10001100
- 00101100

4.

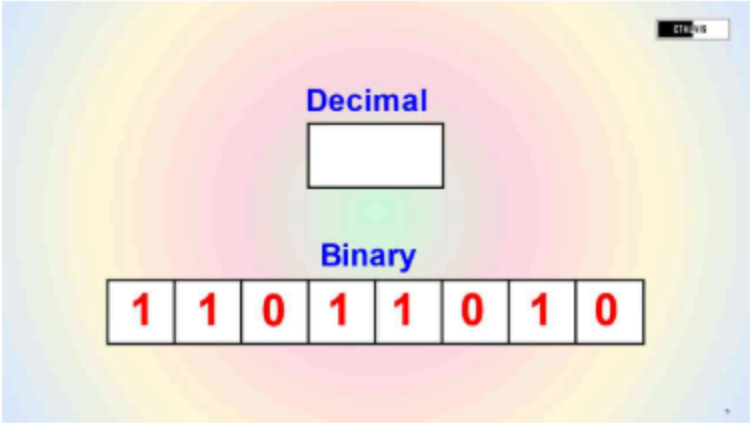


Marks: 1

What is the decimal value?

- 92
- 88
- 90 ✓
- 160

5.

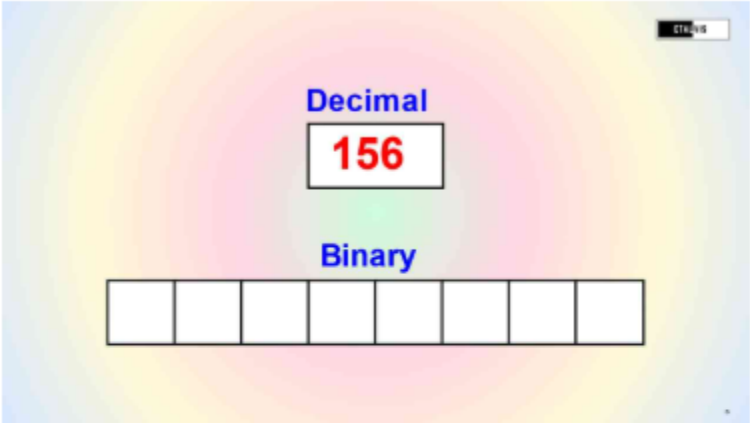


Marks: 1

What is the Decimal Value?

- 218 ✓
- 136
- 98
- 216

6.

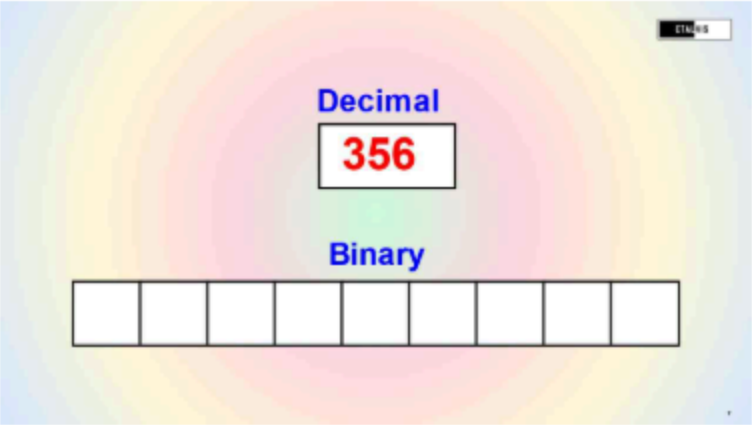


Marks: 1

What is the binary value?

- 01001110
- 10011100 ✓
- 10001110
- 00111110

7.

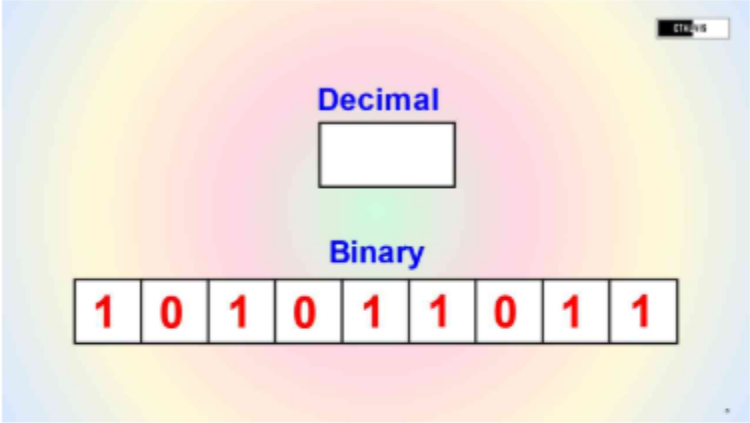


Marks: 1

What is the Binary Value?

- 11001100
- 111001100
- 101100100 ✓
- 01100110

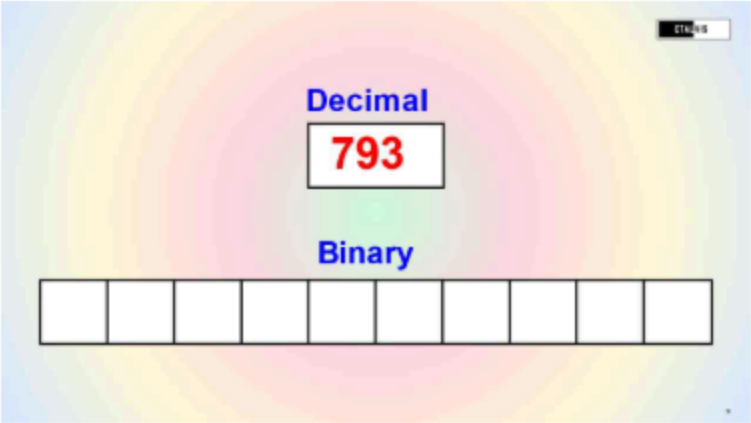
8.



What is the Decimal Value?

- 347 ✓
- 357
- 256
- 197

9.

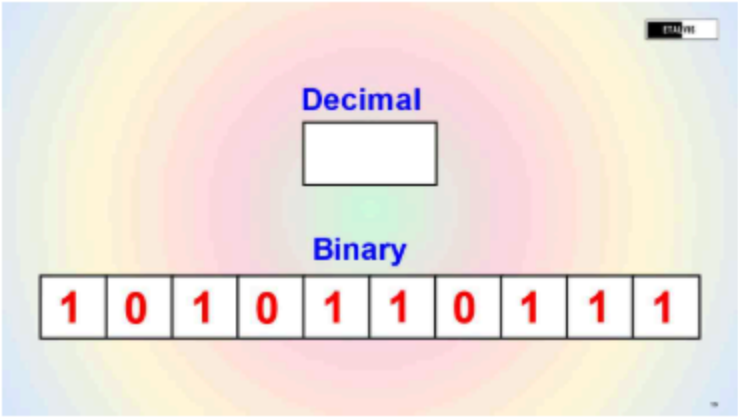


Marks: 1

What is the BInary Value?

- 11101110
- 11001110
- 111001110
- 1100011001 ✓

10.



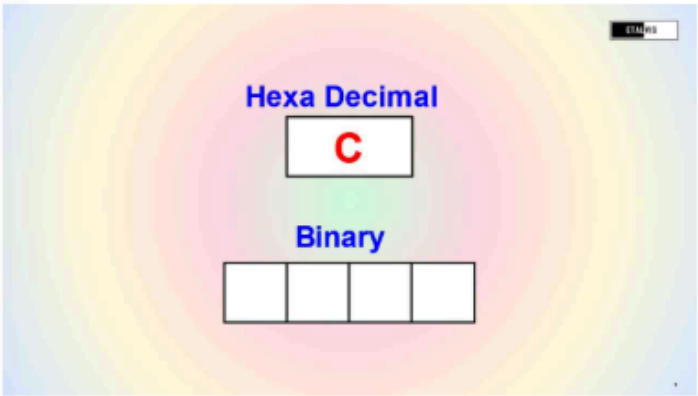
Marks: 1

What is the Decimal Value

- 595
- 695 ✓
- 295
- 395

Assessment:3

1.

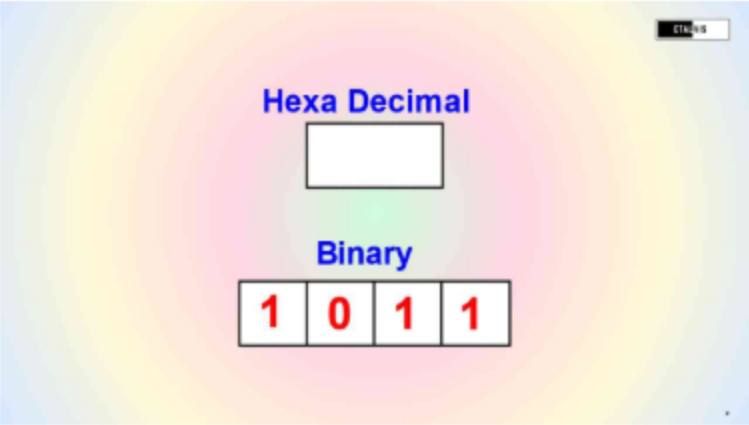


Marks: 1

What is the Binary Value?

- 1010
- 1100 ✓
- 1000
- 0110

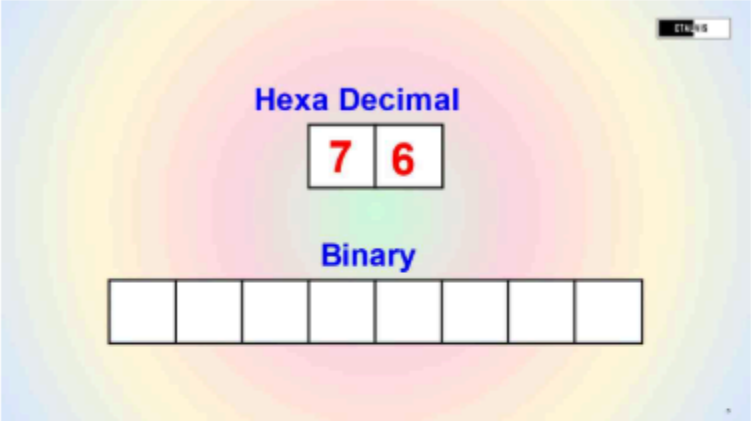
2.



What is the Hexa Decimal Value?

- A
- B
- C
- D

3.

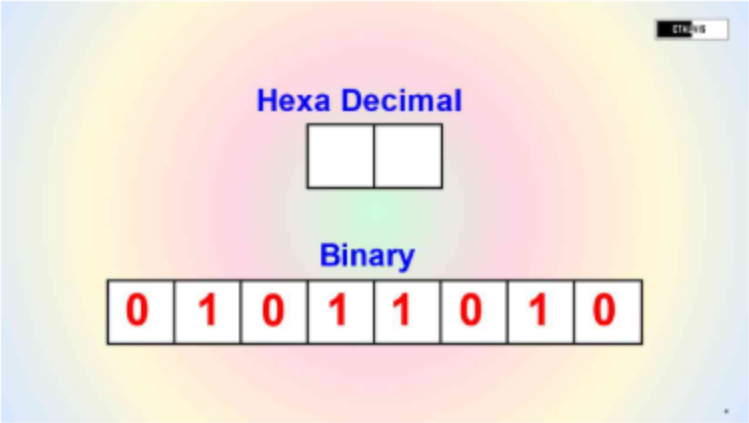


Marks: 1

What is the Binary Value?

- 00110110
- 11000110
- 01111100
- 01110110 ✓

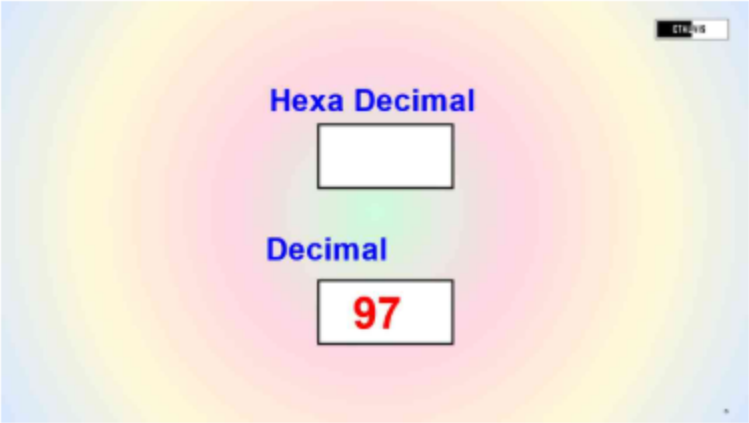
4.



What is the Hexa Decimal Value?

- 5A ✓
- 5B
- 5C
- 5D

6.



What is the HexaDecimal Value?

59	5D
61 ✓	63

7. W

Marks: 1

What is the Octal Value?

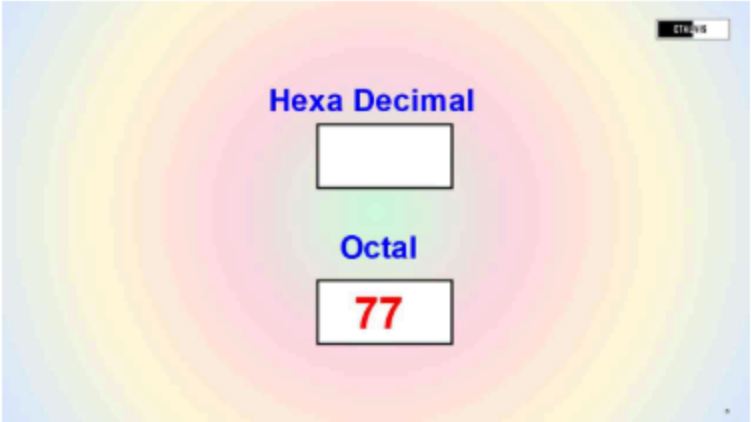
225

227 ✓

229

231

8.

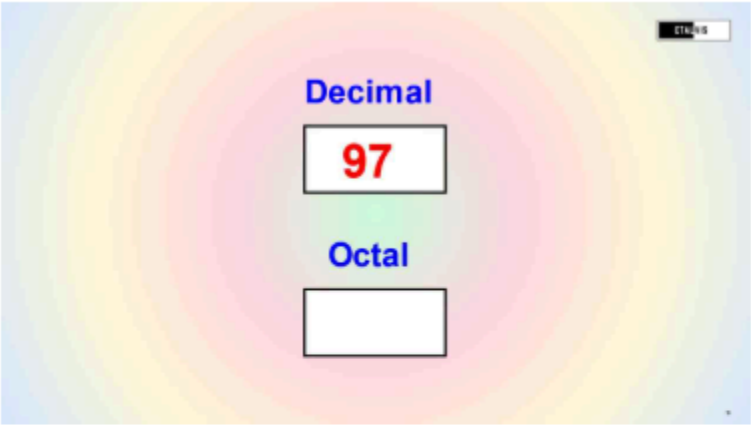


Marks: 1

What is the Hexa Decimal Value

- 3F ✓
- 33
- 77
- 37

9.

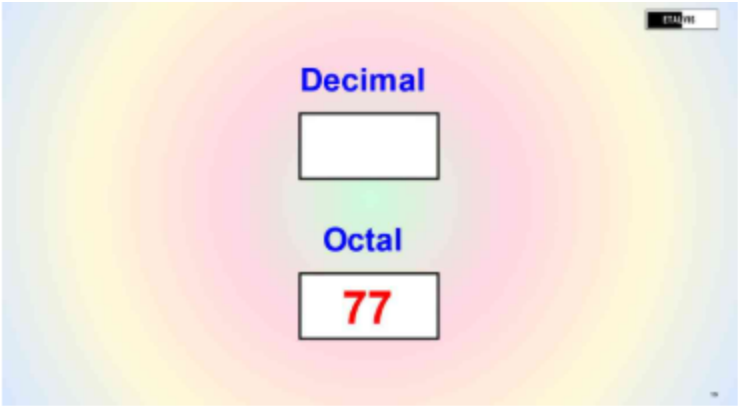


Marks: 1

What is the Octal Calue?

- 139
- 135
- 141 ✓
- 143

10.



Marks: 1

What is the Decimal Value?

- 57
- 63 ✓
- 67
- 71

Assessment:4

1. 2^0=1.
2. 2^1=2.
3. 2^2=4.
4. 2^3=8.
5. 2^4=16.
6. 2^5=32.
7. 2^6=64.
8. 2^7=128.
9. 2^8=256.
10. 2^9=512.

Assessment:5

1. 2^10=1K.
2. 2^20=1M.
3. 2^30=1G.
4. 2^40=1T.
5. 2^50=1P.
6. 2^12=4K.
7. 2^24=16M.
8. 2^36=64G.
9. 2^48=256T.
10. 2^53=8P.

Assessment:6

What is the value of X?

1. 2^X=128K.
2. 2^X=256M.
3. 2^X=64G.
4. 2^X=32T.
5. 2^X=16P.
6. 2^X=4G.
7. 2^X=8M.
8. 2^X=2K.
9. 2^X=512T.
10. 2^X=64P.

Assessment:7

1.

0	1	0	0
+ 1	1	0	1

Marks: 1

Choose the correct answer.

- 1101
- 10001 ✓
- 1001
- 1101

2.

0	1	1	0	1	1	0	0
+ 1	0	1	1	0	1	0	1

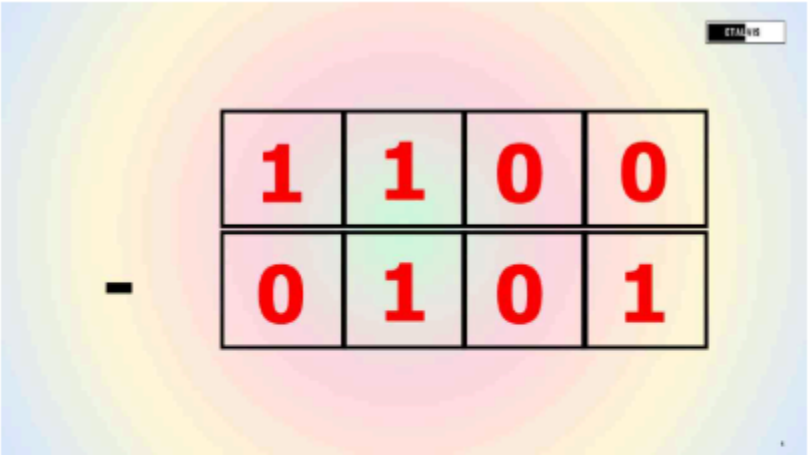
Marks: 1

Choose the correct answer

- 110100001
- 100101001
- 100100001 ✓
- 101100001

Assessment:8

1.

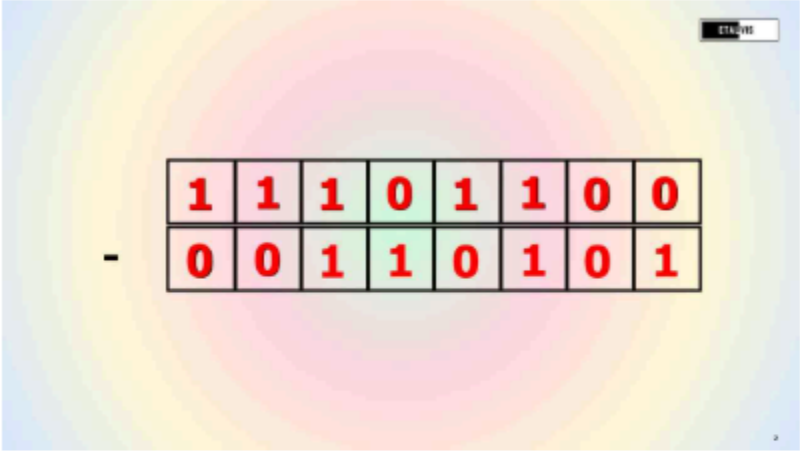


Marks: 1

Choose the correct answer.

- 0011
- 0111 ✓
- 1011
- 0001

2.



Marks: 1

Choose the correct answer.

- 10110111 ✓
- 10110011
- 10100111
- 11000111

3.

Marks: 1

	1	1	1	0	1	1	0	0
OR	0	0	1	1	0	1	0	1

Choose the correct Answer

- 11111101 ✓
- 00100100
- 11111111
- 11011001

4.

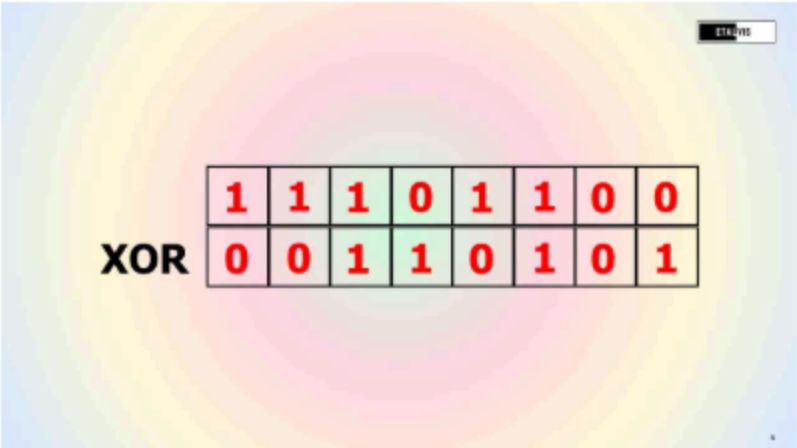
Marks: 1

NOT	1	1	1	0	1	1	0	0

Choose the correct answer.

- 00110011
- 00010001
- 00010011 ✓
- 00010000

5.



Marks: 1

Choose the correct answer.

- 11111101
- 00100100
- 11110100
- 11011011
- 11011001 ✓
- None of the above

6.

Marks: 1

Choose the correct answer.

- 00010011 ✓
- 11100011
- 00011100
- 11111111

7.

Marks: 1

Choose the correct answer.

- 00010011
- 00010100 ✓
- 00010101
- 10010100
- None of the above

8.

Marks: 1

	1	0	1	0	1	1	0	0
-	1	1	1	1	0	1	0	1

Choose the correct answer.

- -10110111 ✓
- 10110111 ✗
- -01111011
- 01111011

9.

	1	1	1	0	1	1	0	0
AND	0	0	1	1	0	1	0	1

Choose the correct Answer

- 00000100
- 00110100
- 00100100 ✓
- 00100101

Assessment:8

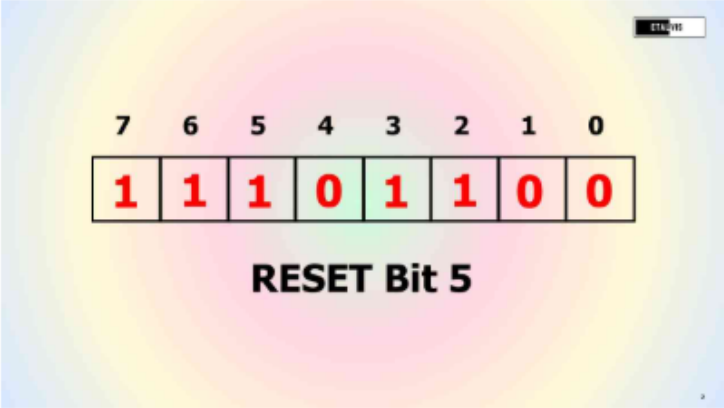
1.

Marks: 1

Which Operation to be performed to SET a bit in a binary number

- AND
- OR ✓
- NOT
- XOR

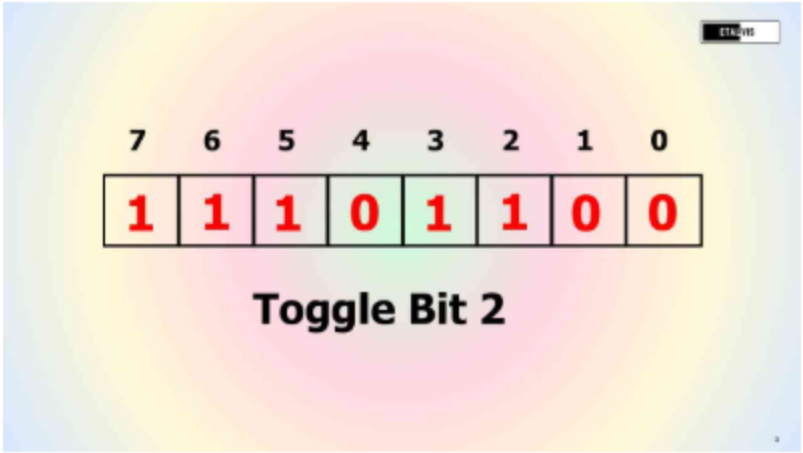
2.



Which Operation to be performed to RESET a bit in a binary number

- OR
- AND
- NOT
- XOR

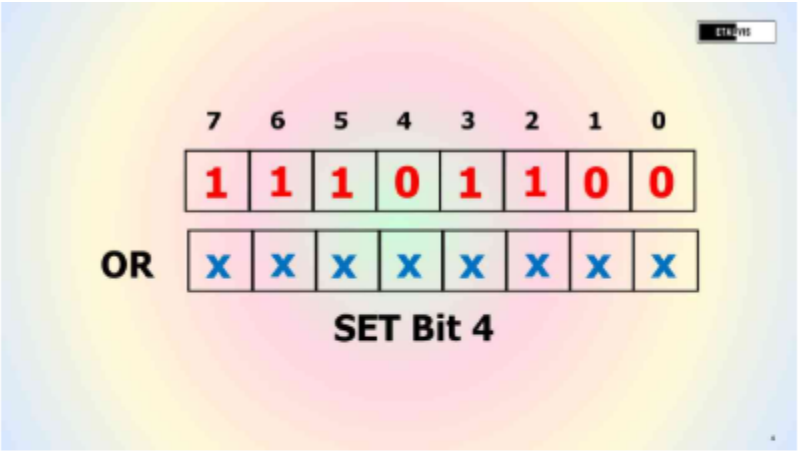
3.



Which Operation to be performed to Toggle a bit in a binary number

- OR
- AND
- NOT
- XOR

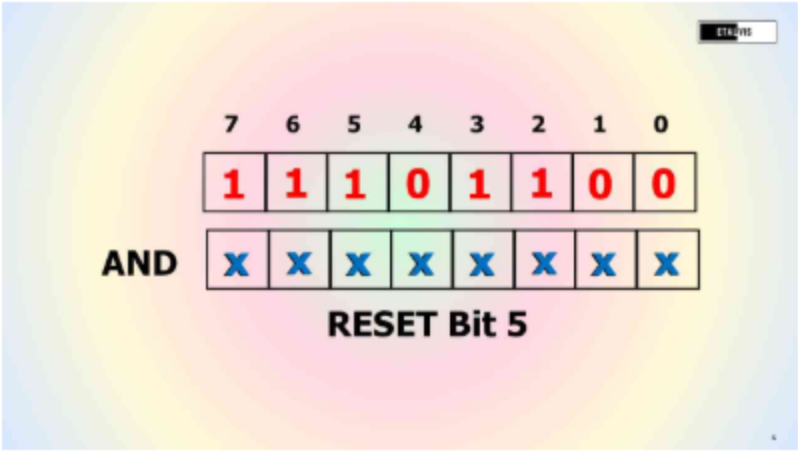
4.



Find the Binary value to SET bit 4

- 11111111
- 11101111
- 00010000 ✓
- 00001000

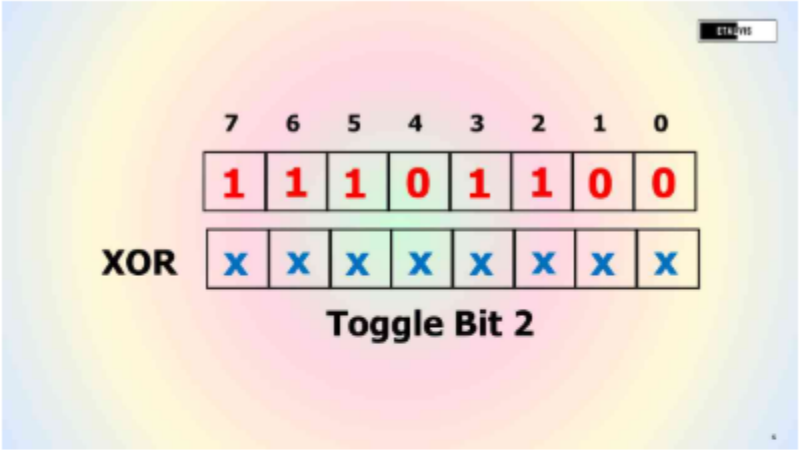
5.



Find the Binary value to RESET bit 5

- 11011111 ✓
- 00100000
- 11111111
- 00000000

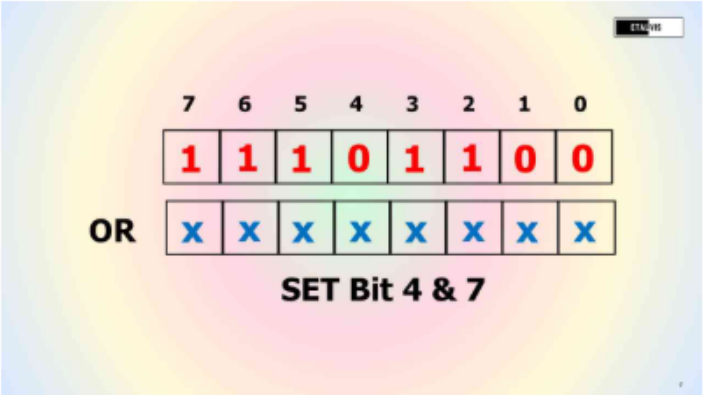
6.



Find the Binary value to Toggle bit 2

- 11111011
- 11101000
- 00000100 ✓
- 00000010

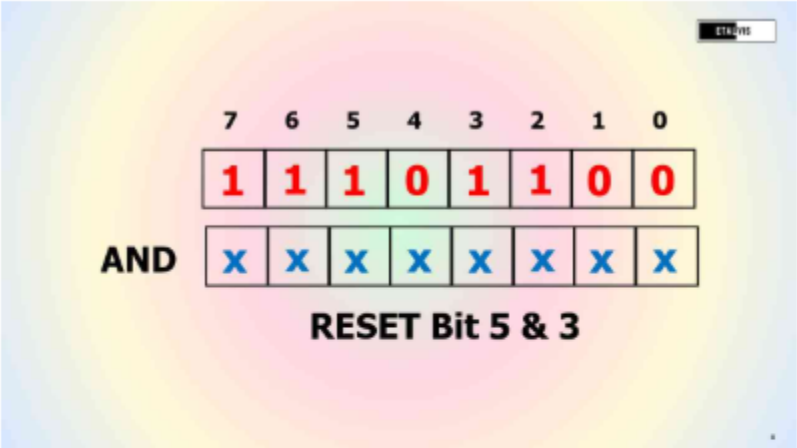
7.



Find the Binary value to SET bits 4 & 7

- 11111100
- 10010000
- 11111111
- 00010000
- None of the above

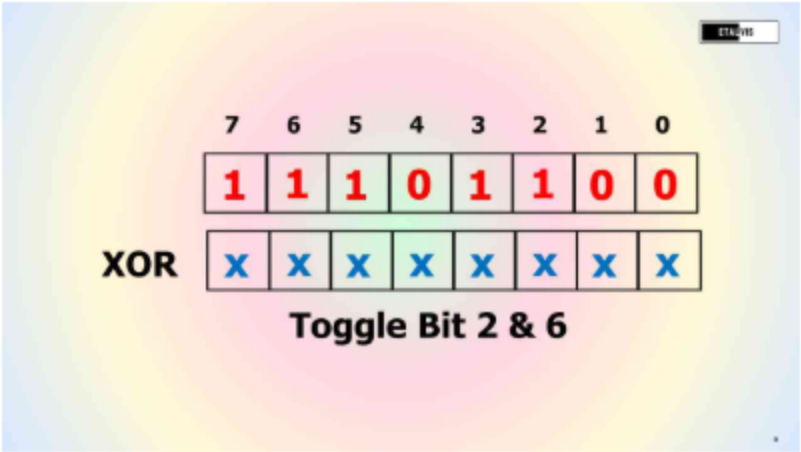
8.



Find the Binary value to RESET bits 3 & 5

- 11000100
- 00000000
- 11010111 ✓
- 11111111

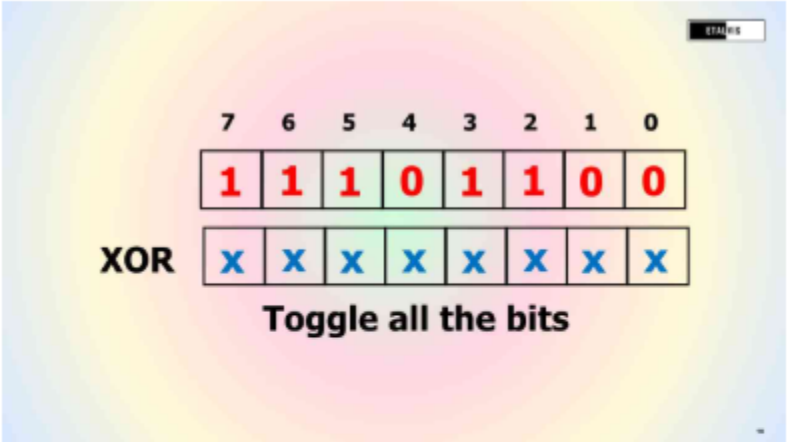
9.



Find the Binary value to TOGGLE bits 2 & 6

10101000	01010111
11111111	01000100

10.



Find the Binary value to TOGGLE all bits

00010011	11101100
11111111 ✓	00000000

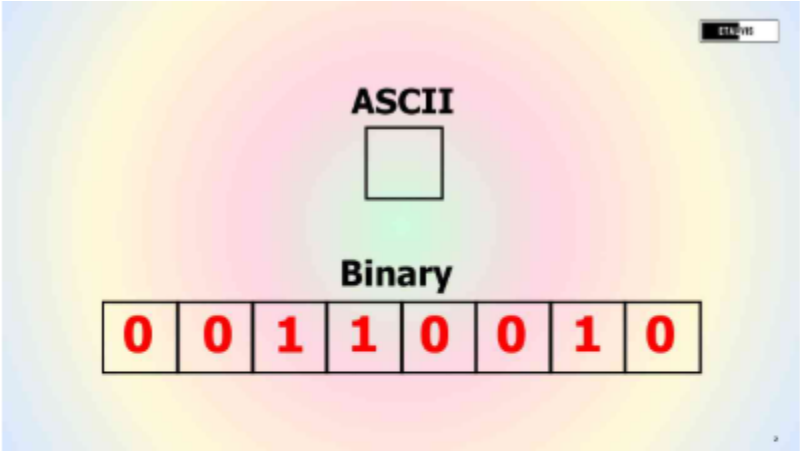
Assessment:9

1.

Choose the Correct Answer.

- 00000000 ✕
- 00110000
- 00110010
- 00110001

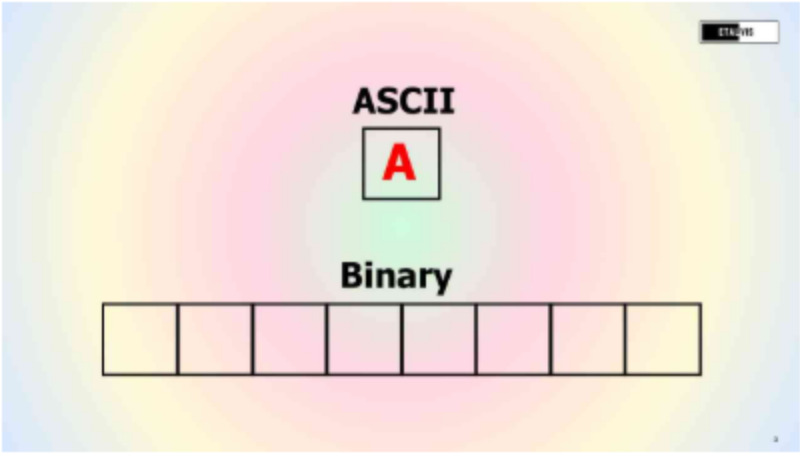
2.



Choose the correct answer.

- 1
- 2
- 3
- 4

3.



Choose the correct answer.

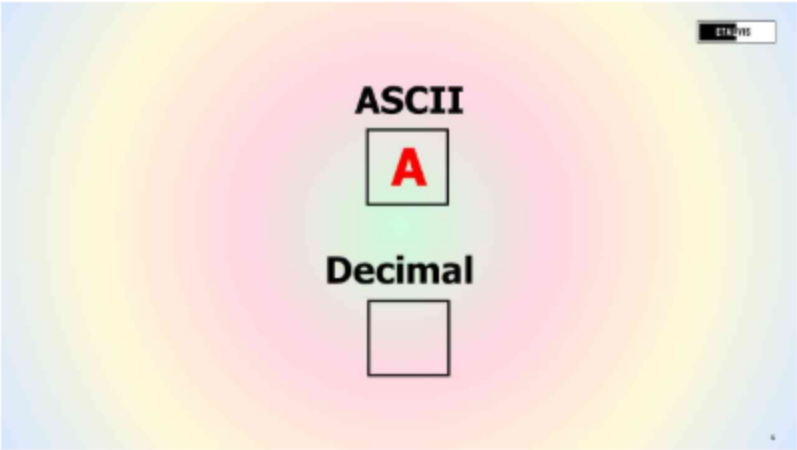
- 01000001 ✓
- 01001010
- 10101010
- 01100001

4.

Choose the correct answer.

- D
- E
- F ✓
- G

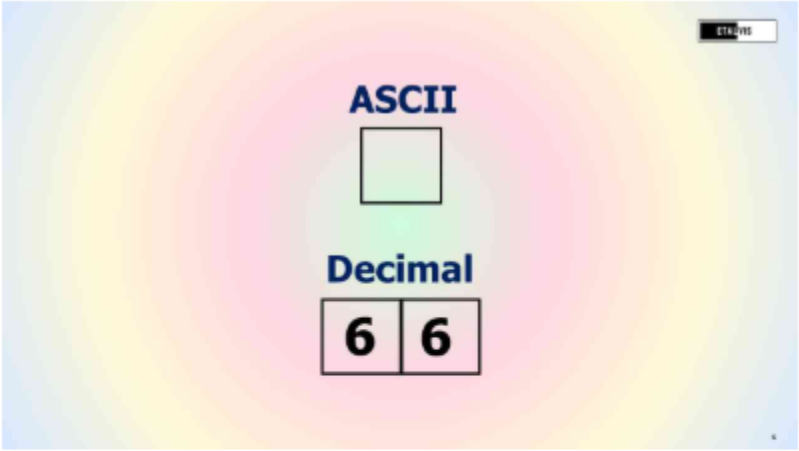
5.



Choose the correct answer.

- 41
- 49
- 61
- 65

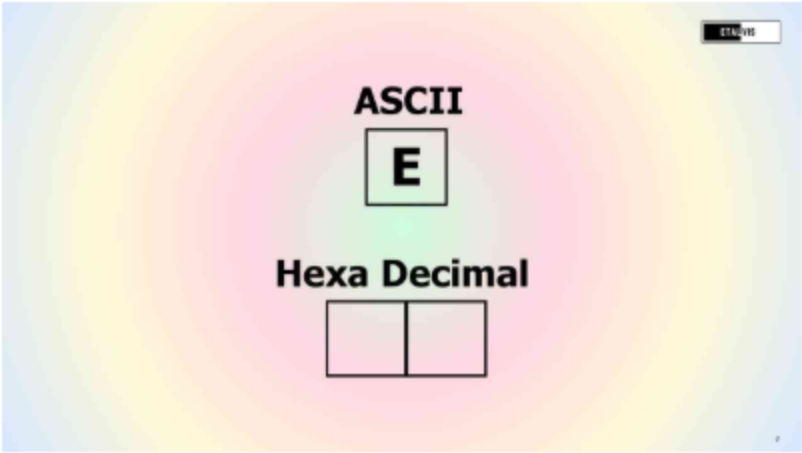
6.



Choose the correct answer.

- A
- a
- B ✓
- b

7.



Choose the correct answer.

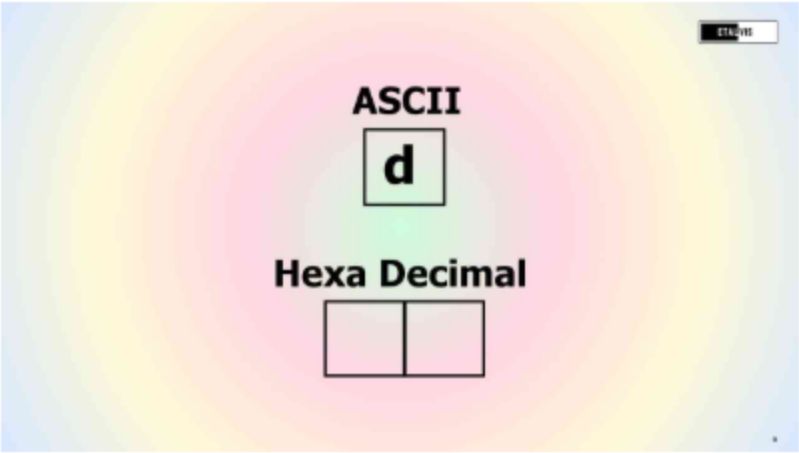
- 41
- 42
- 44
- 45

8.

Choose the correct answer.

- E
- F
- G
- H

9.



Choose the correct answer.

- 63
- 64 ✓

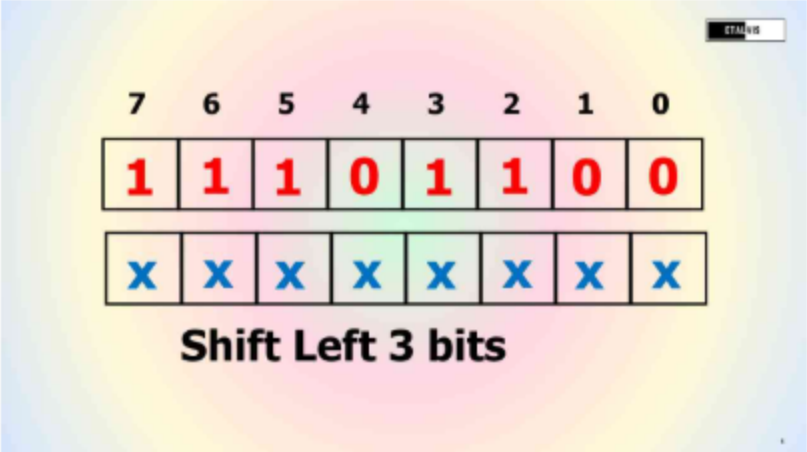
10.

Choose the correct answer.

- a
- d
- f ✓
- h

Assessment:10

1.



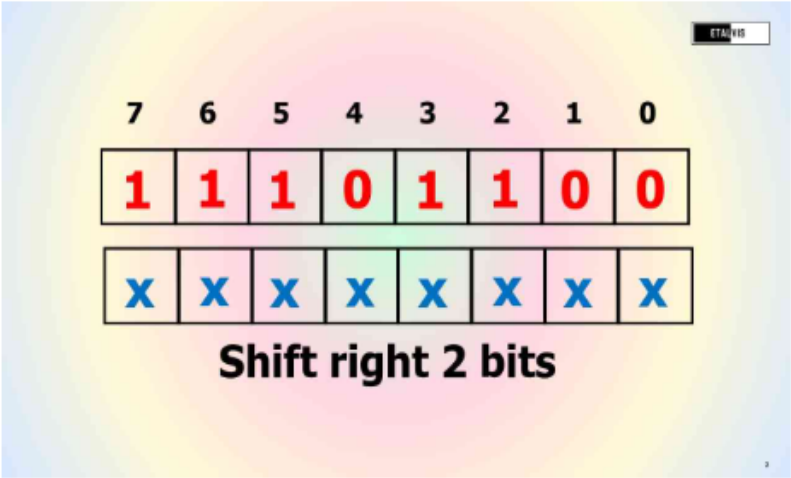
Mc

What is the binary value after shifting the bits left?

01100000 ✓

01100001

2.



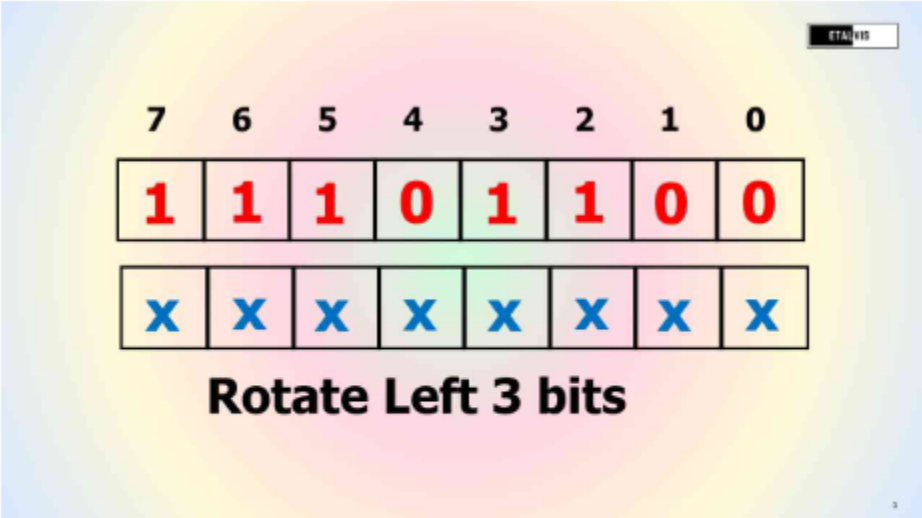
Marks:

What is the binary value after shifting the bits left?

11111011

00111011

3.

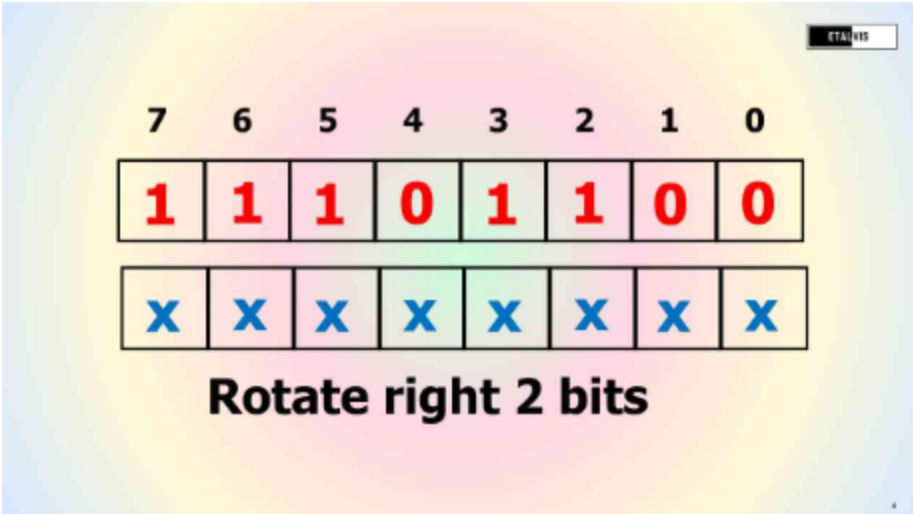


What is the binary value after rotating the bits left

01100000

01100111

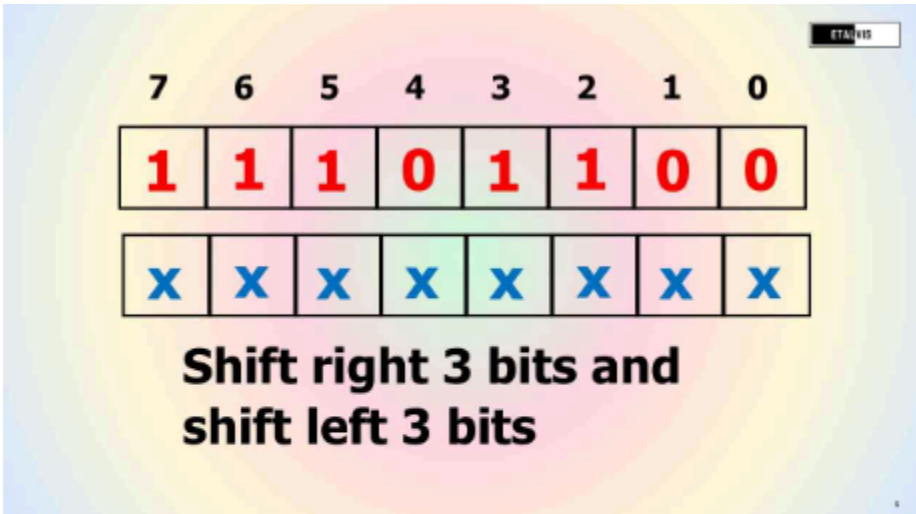
4.



What is the binary value after rotating the bits right

- 00111011 ✓
- 00111000

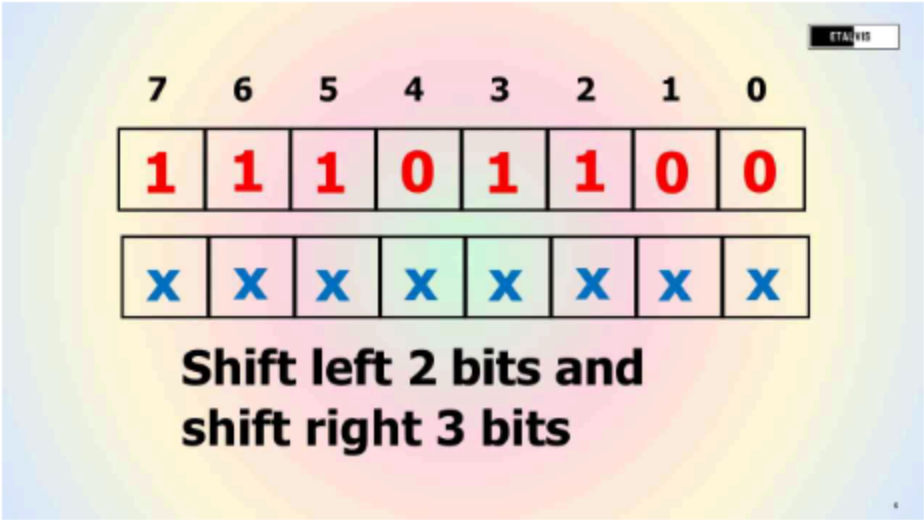
5.



What is the binary value after shifting the bits right & left

- 11101111
- 11101100
- 11101000 ✓
- 00001000

6.

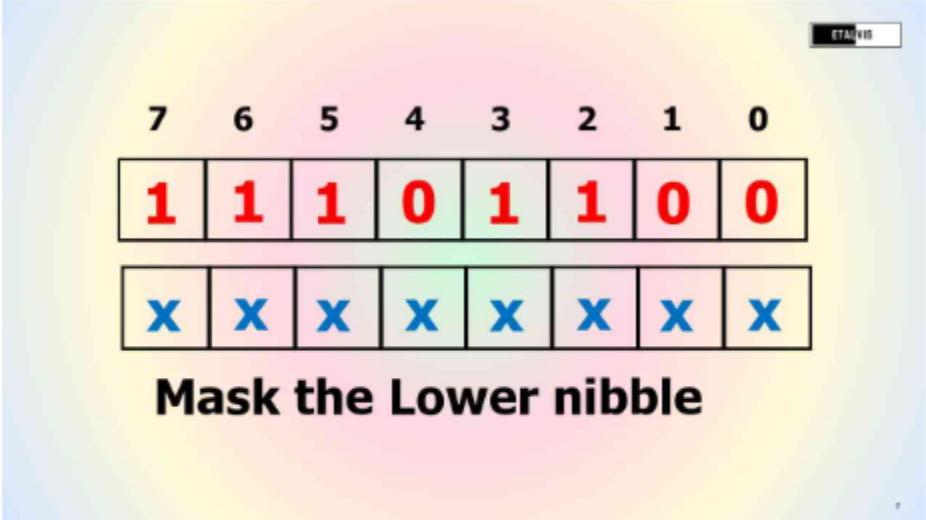


Marks: 1

What is the binary value after shifting the bits left & right

- 11110110
- 01110110
- 0110110
- 00010110 ✓

7.

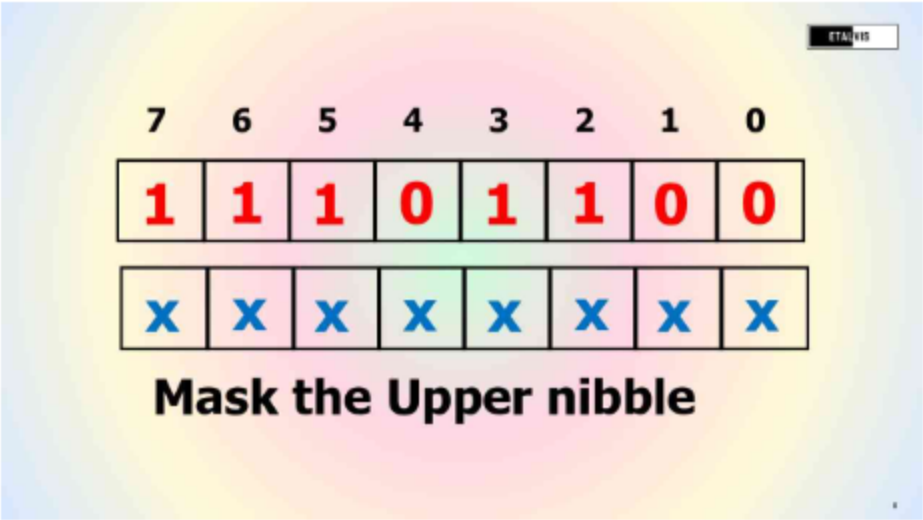


Marks:

What is the binary value to mask the lower nibble

- 00001111
- 11110000 ✓

8.



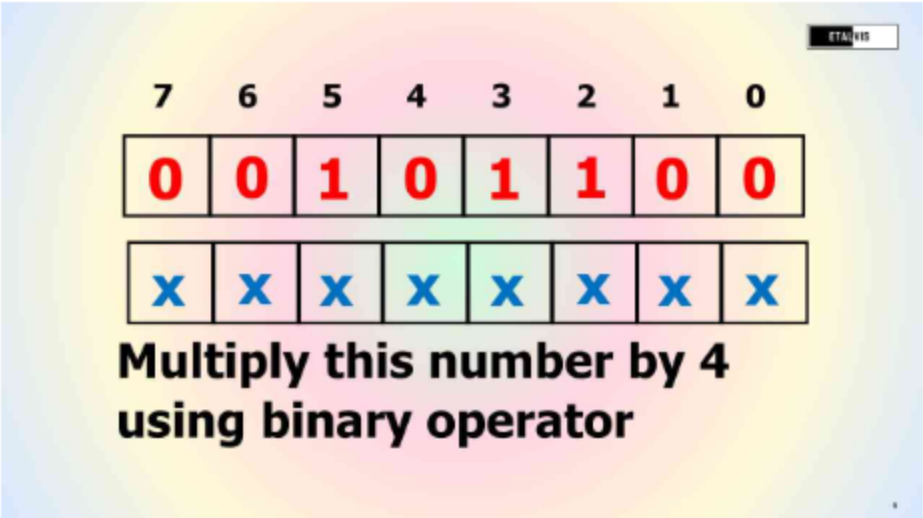
Mar

What is the binary value to mask the Upper nibble

00001111 ✓

11110000

9.



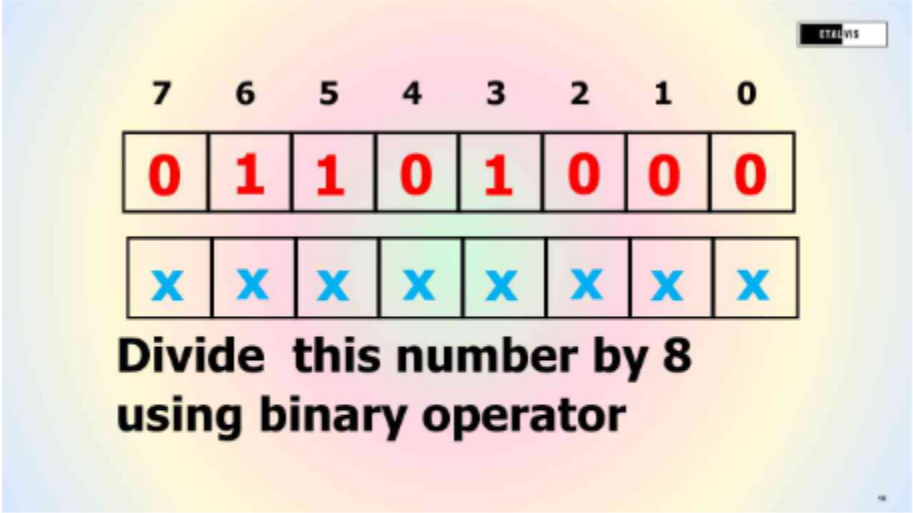
Mc

What is the binaryoperatorr to use to multiply by 4

Left Shift ✓

Right Shift

10.



Marks: 1

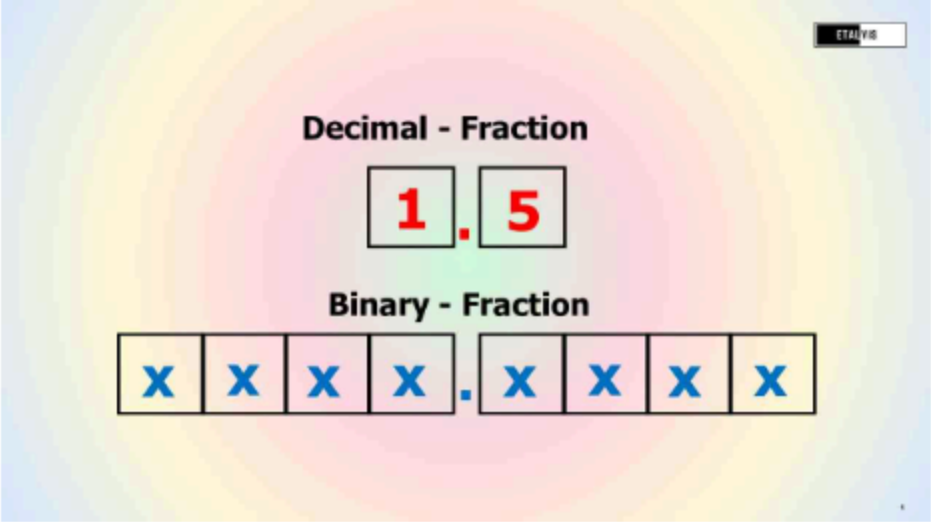
What is the binary operator to use to divide by 4

Left Shift

Right Shi... ✓

Assessment:11

1.

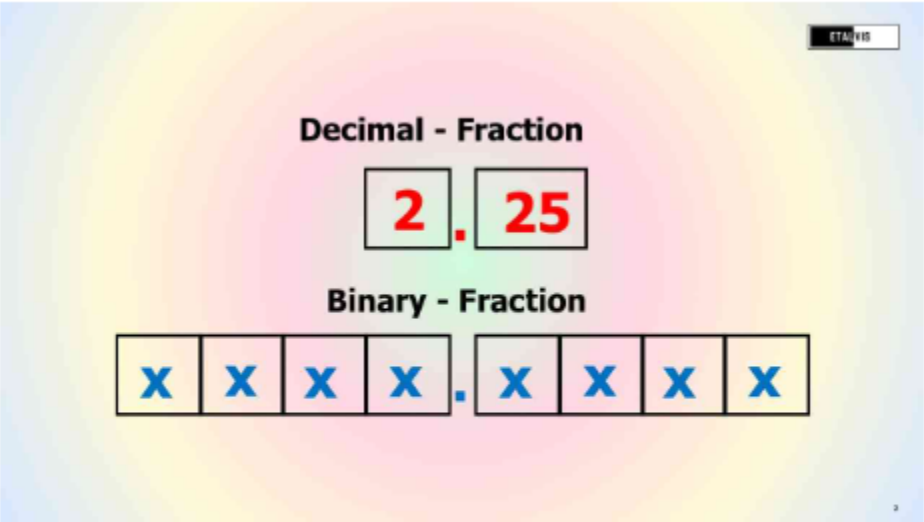


Mc

What is the Binary Fraction?

- 1111.1111
- 0000.1111
- 0001.1000 ✓
- 0010.0100

2.

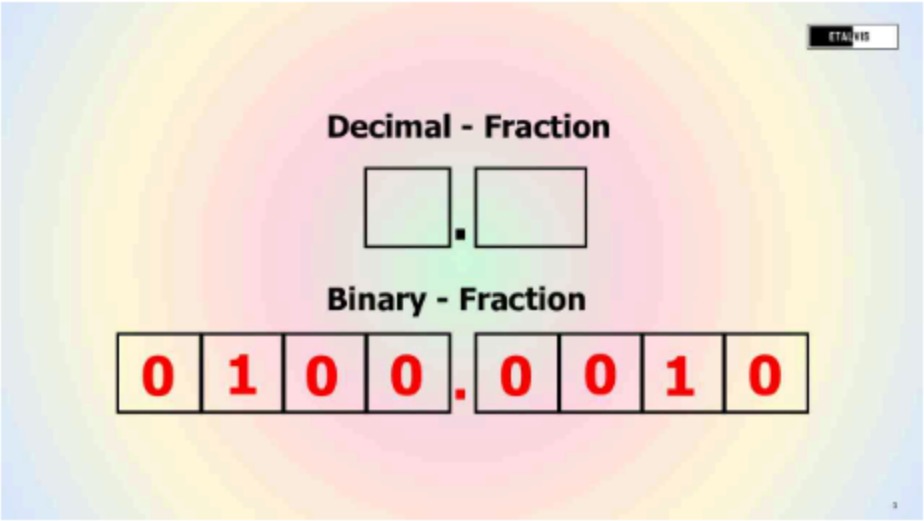


Marks: 1

What is the Binary Fraction?

- 0010.1000
- 0010.0100 ✓

3.

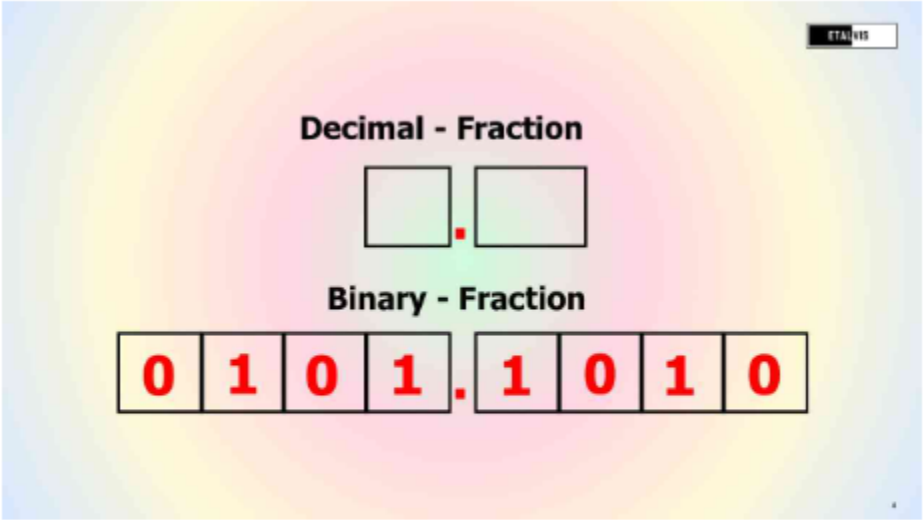


Mc

What is the Value of Decimal Fraction?

- 4.5
- 4.25
- 4.125 ✓
- 4.0625

4.

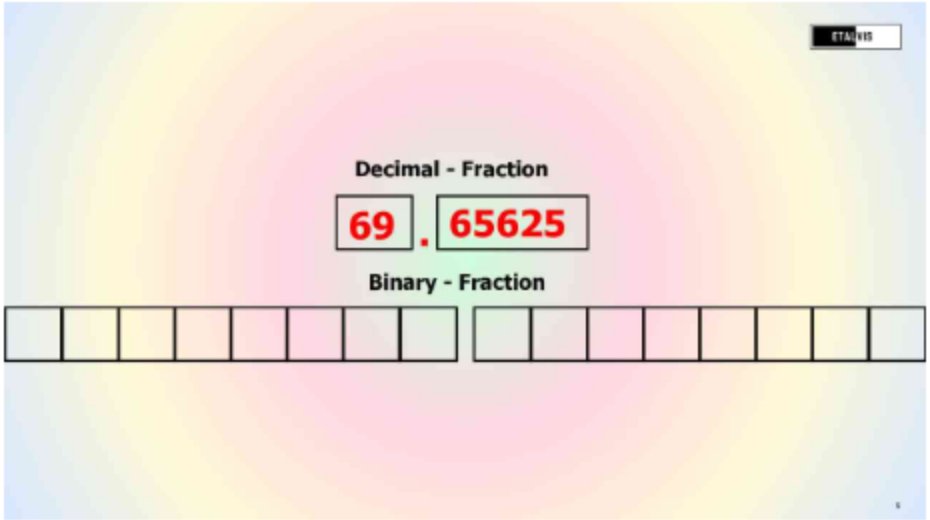


Marks:

What is the Decimal Fraction?

- 5.0625
- 5.625 ✓

5.



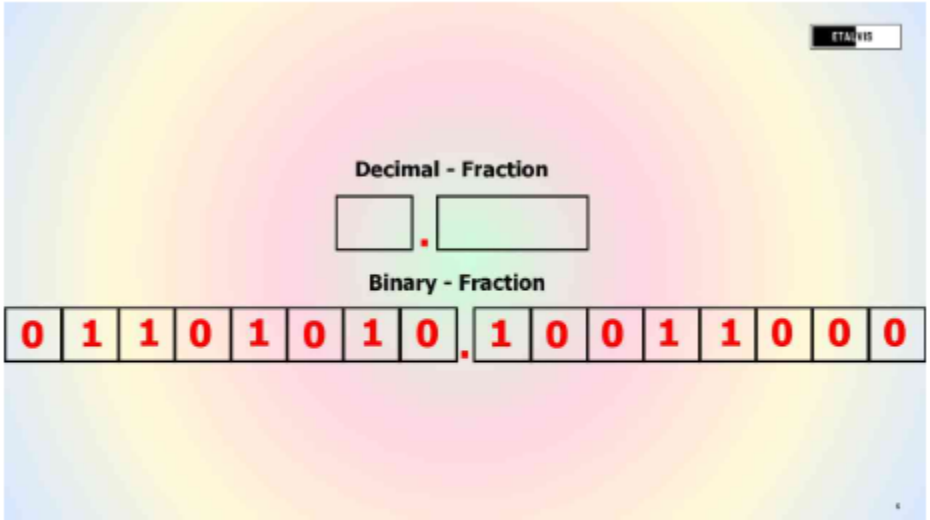
Mark

What is the Binary Fraction?

01000101.10101000 ✓

01000101.10001000

6.



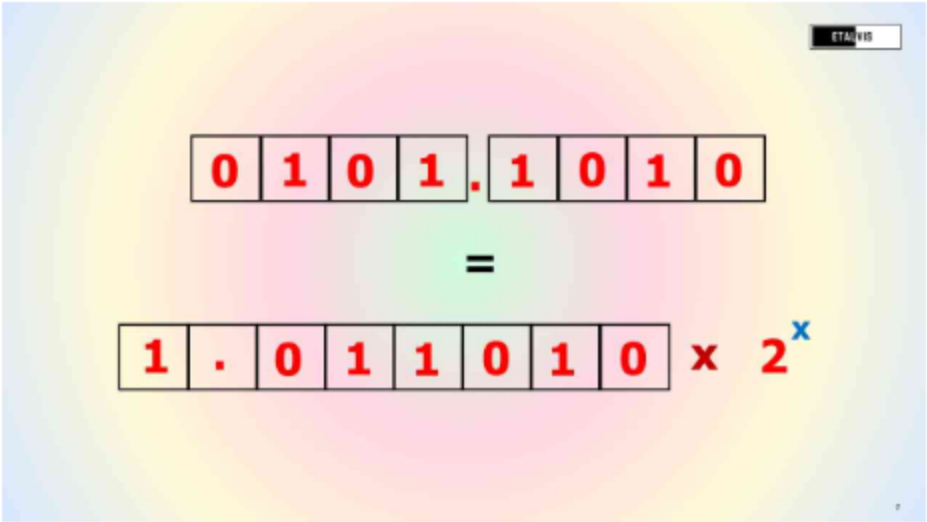
Mar

What is the Decimal Fraction?

106.59375 ✓

106.125

7.



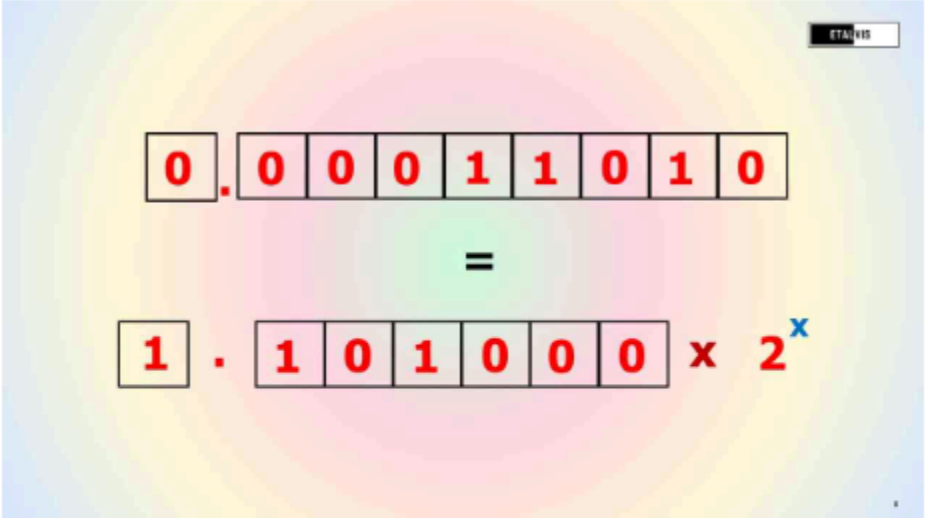
Marks: 1

What is the value of 'x'

1

2 ✓

8.



Marks

What is the value of 'x'

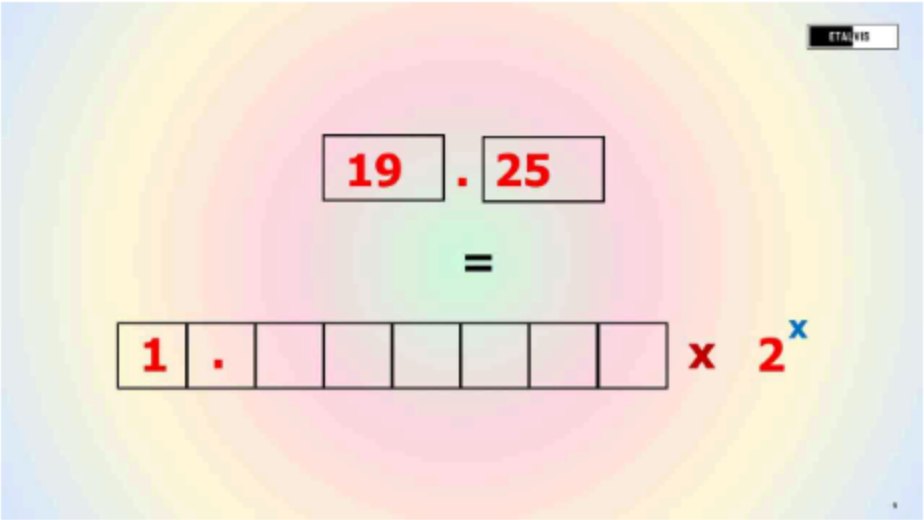
-1

-2

-3

-4

9.

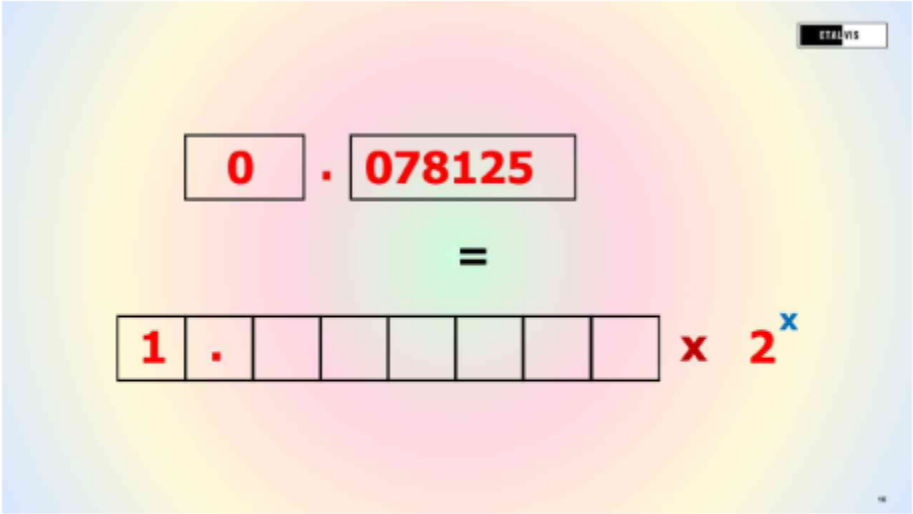


Marks: 1

What is the value of 'x'

1	2
3 ✗	4 ✓

10.



Marks: 1

What is the value of 'x'

-3	-4 ✓